Module 6 – Final Project Whitepaper

Connecticut Property Sales (1999-2023)

Tableau Visualization

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Abstract:

This project analyzes real estate sales data from 2001 to 2022, providing insights into property transactions across various towns. The study aims to uncover trends in property sales, price variations, and assessment accuracy using data visualization techniques in Tableau. Through interactive dashboards, we will explore the relationship between assessed values and sale prices, geographical sales distribution, and property type preferences. The findings will assist real estate investors, policymakers, and buyers in making informed decisions.

Introduction:

Real estate markets are dynamic and influenced by multiple factors, including property type, location, assessed value, and market demand. Understanding these factors through historical sales data can help identify patterns and predict future trends. This study utilizes a dataset comprising over 1 million property sales records to visualize key real estate metrics.

Data Overview:

- The dataset consists of 1,048,575 records with the following key attributes:
- Transaction Details: Serial Number, List Year, Date Recorded, Town, Address
- Financial Metrics: Assessed Value, Sale Amount, Sales Ratio
- Property Classification: Property Type (Residential, Commercial), Residential Type (Single Family, Condo, etc.)
- Geographic Data: Latitude, Longitude (available for ~30% of records)

Preprocessing Steps:

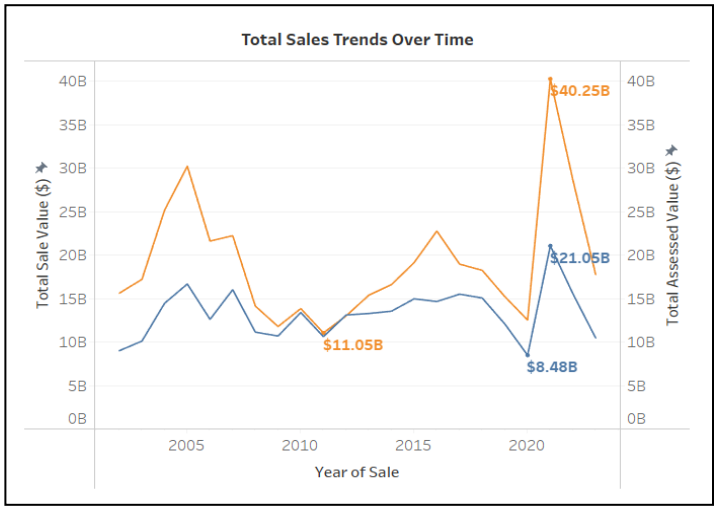
- Converted blank values to NA values
- Replaced NA values in categorical columns using mode.
- Dropped columns with more than 60% missing data.

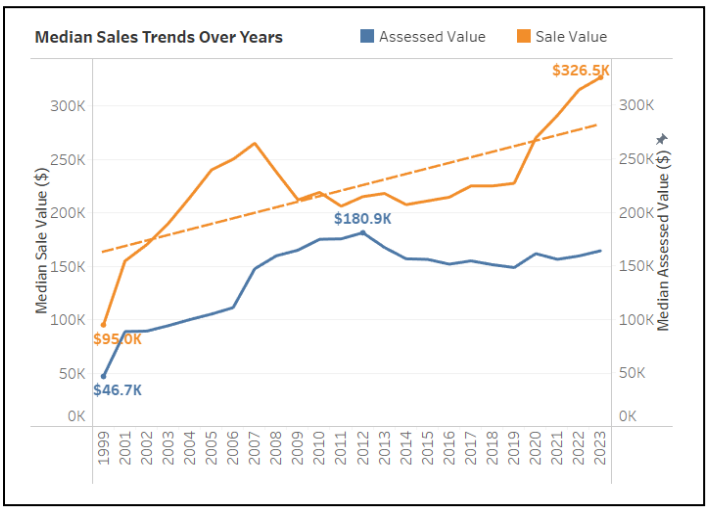
Research Questions for Tableau Dashboard:

1. How have real estate sales trends changed over time?
2. Is there a seasonal trend in property sales?
3. Which towns have shown the highest property price growth over time?
4. How has property values appreciated year-over-year across different property types?
5. What is the distribution of median sales between different residential types?
6. What is the distribution of median sales between different property types?

Tableau Visualizations

Sale Trends over Years (Sale Value vs. Assessed Value)

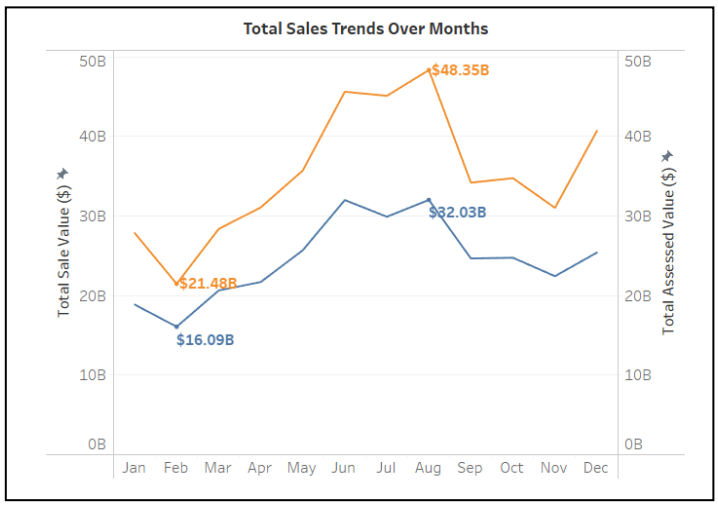


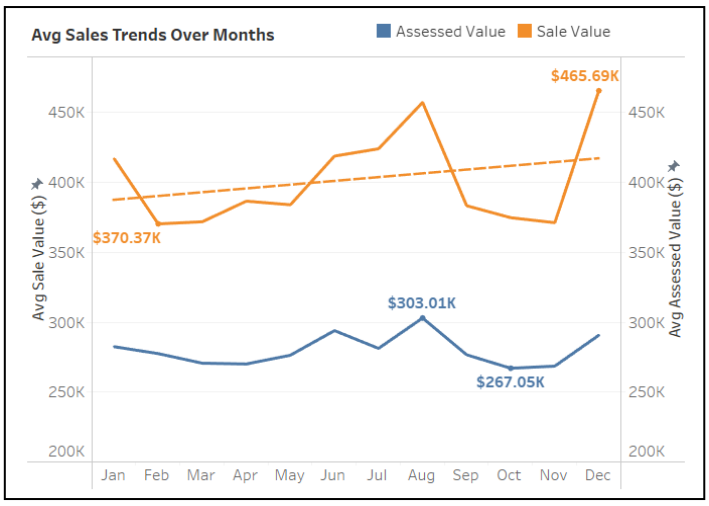


Insights:

- The 2008 financial crisis caused a sharp decline in sales, while 2020-2021 saw a massive surge (~$40.25B) due to low interest rates and pandemic-driven demand.
- Median sale value rose from $95K (1999) to $326.5K (2023), consistently outpacing assessed value growth, indicating potential under-assessment or market inflation.
- The decline after 2021 suggests a cooling real estate market, likely due to rising interest rates and economic adjustments.

Sale Trends over Months (Sale Value vs. Assessed Value)

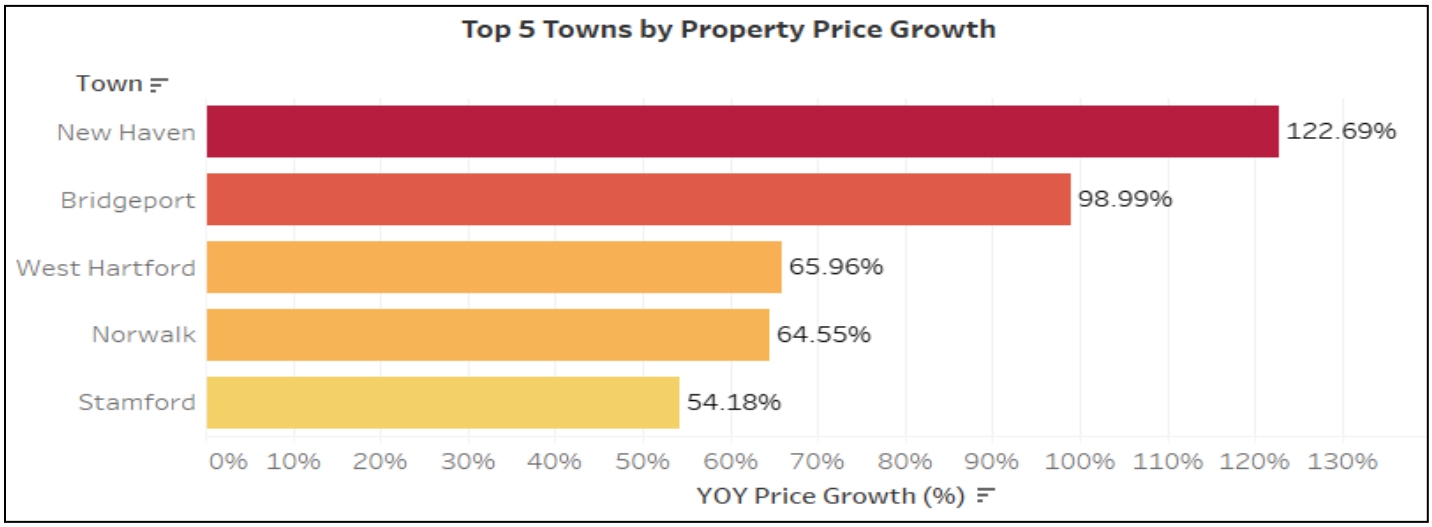




Insights:

Spring & summer months (March–August) show peak activity, while early winter (January-February) and post-summer (September-October) experience slowdowns.

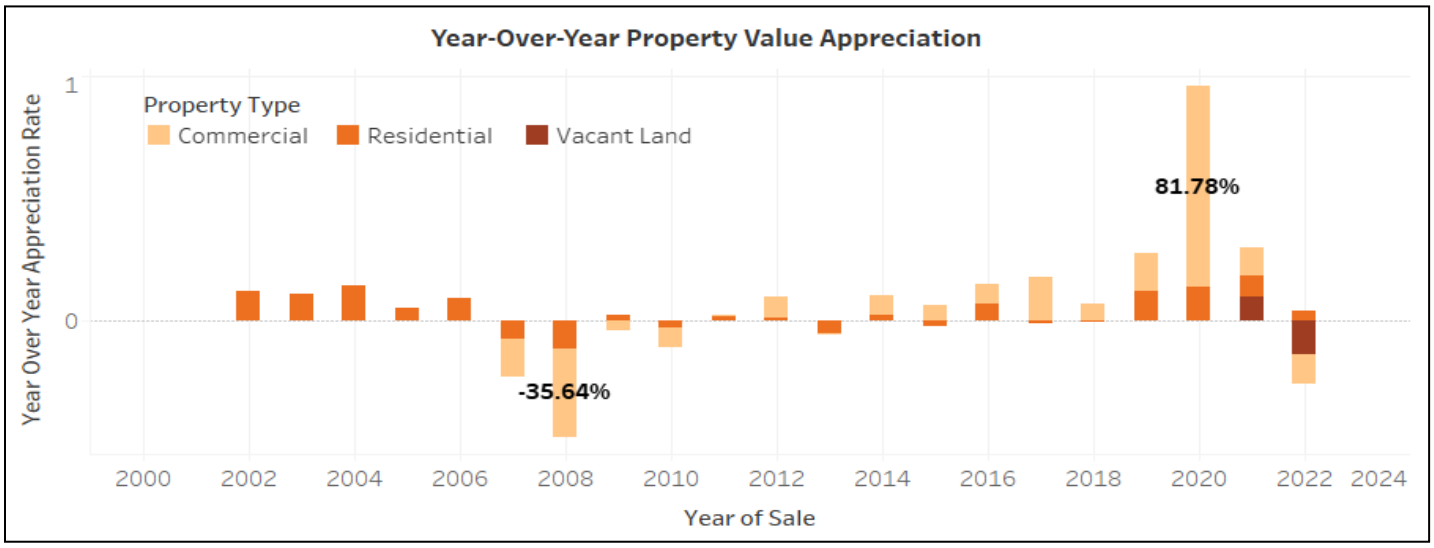
Top 5 Towns by Year-Over-Year Property Price Growth



Insights:

- New Haven leads with the highest YoY price growth (122.69%), followed by Bridgeport (98.99%), indicating strong real estate demand in these areas. West Hartford, Norwalk, and Stamford also experienced significant price appreciation (54%-66%), making them attractive markets for investors and homeowners.
- The high growth rates suggest increasing property values, possibly driven by economic development, rising demand, or improved infrastructure in these towns.

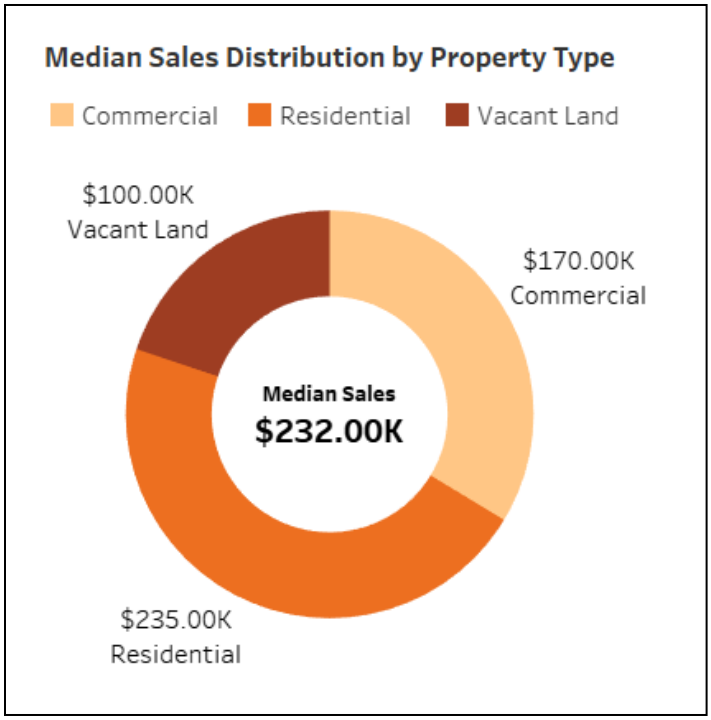
Year-Over-Year Property Value Appreciation



Insights:

- 2008 Financial Crisis Impact: A sharp decline (-35.64%) in 2008 indicates a major drop in property values due to the global financial crisis, affecting all property types.
- Pandemic Boom (2020): A massive appreciation (81.78%) in 2020 suggests increased demand for real estate, likely driven by low interest rates and shifting housing needs during COVID-19.
- Post-2021 Market Correction: A decline in property appreciation after 2021 suggests a cooling market, likely due to rising interest rates and economic adjustments.

Median Sale Distribution by Property Type



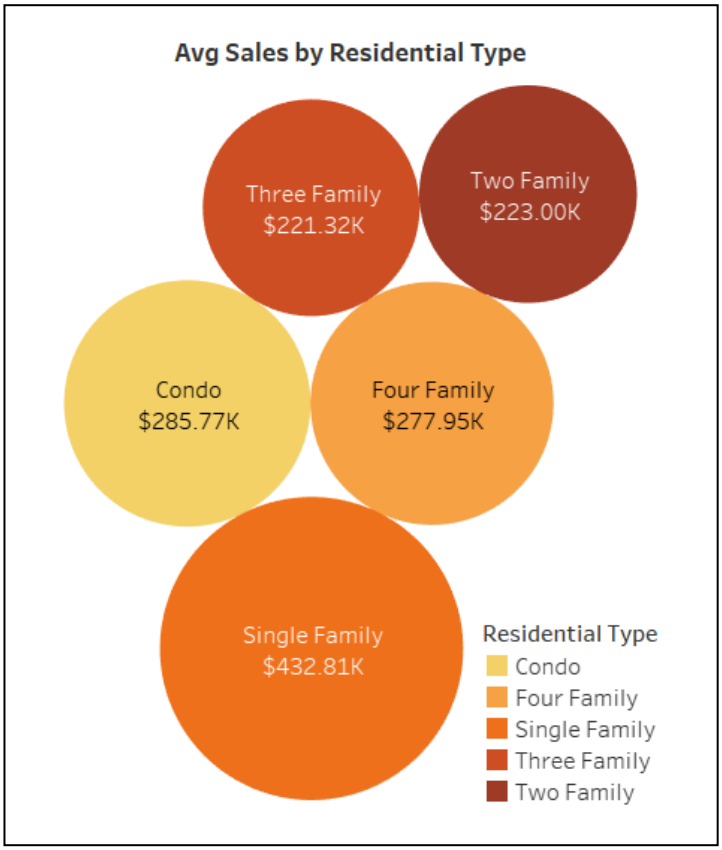
Insights:

Residential properties have the highest median sale value ($235K), indicating strong demand and higher valuation compared to other property types.

Commercial properties have a median sale value of $170K, suggesting moderate demand, likely influenced by economic conditions and business investments.

Vacant land has the lowest median sale value ($100K), reflecting lower market valuation, possibly due to limited immediate usability or development potential.

Median Sale Distribution by Residential Type



Insights:

Single Family homes have the highest average sale price ($432.81K), indicating strong demand and higher valuation compared to other residential types.

Condos and Four Family homes have moderate average prices ($285.77K and $277.95K, respectively), suggesting they offer affordability while still holding value in the market.

Two Family and Three Family homes have the lowest average prices ($223K and $221.32K), making them more accessible for buyers, potentially appealing to investors or rental property owners.

Key Findings:

1. Market Trends & Fluctuations

- 2008 Financial Crisis Impact: A sharp decline in total sales value (-35.64%) during 2008-2011 indicates a major real estate downturn.
- Pandemic Boom (2020-2021): A massive surge in total sales (~$40.25B in 2021, 81.78% YoY appreciation) was driven by low interest rates and shifting housing demands.
- Post-2021 Market Correction: A sharp decline after 2021 suggests rising interest rates and economic adjustments cooling the market.

2. Property Price Growth & Appreciation

- New Haven (122.69%) and Bridgeport (98.99%) had the highest YoY property price growth, making them attractive for real estate investments.
- West Hartford, Norwalk, and Stamford also saw strong appreciation (54-66%), highlighting consistent market demand.
- Median sales price has increased from $95K (1999) to $326.5K (2023), significantly outpacing assessed value growth.

3. Sale vs. Assessed Value Trends

- Assessed values consistently lag behind sale values, indicating under-assessment or market-driven inflation.
- Single Family homes exhibit the largest gap between sale and assessed value, suggesting strong demand in this category.

4. Property Type Distribution & Valuation

- Residential properties hold the highest median sales value ($235K), followed by commercial ($170K) and vacant land ($100K), suggesting lower demand for undeveloped land.
- Single Family homes have the highest average sales price ($432.81K), while multi-family homes (Two, Three, and Four Family) are more affordable ($221K-$278K), making them attractive for rental investment.

5. Seasonal & Monthly Trends

- Spring & Summer months (March–August) see peak transactions, while Winter (January-February) and Post-Summer (September-October) show slowdowns.

Recommendations:

1. **For Buyers & Investors:**

- Consider towns with a low sales ratio, as properties may be undervalued.
- Look at trends in residential type pricing to make cost-effective decisions.

2. **For Government & Assessors:**

- Improve property assessment accuracy in towns with high discrepancies between and sale price.
- Ensure fair and standardized property evaluations to prevent over/under-taxation.

3. **For Real Estate Agents & Developers:**

- Focus on high-growth towns for new developments.
- Use historical sales data to strategize marketing and pricing.

Conclusion:

The Connecticut real estate market has experienced major fluctuations influenced by economic downturns, recovery phases, and external factors like the pandemic. Residential properties, especially Single Family homes, remain the most valuable, while certain towns have shown high appreciation, making them investment hotspots. However, recent market corrections indicate potential cooling, making it essential for investors and policymakers to analyze future trends cautiously.

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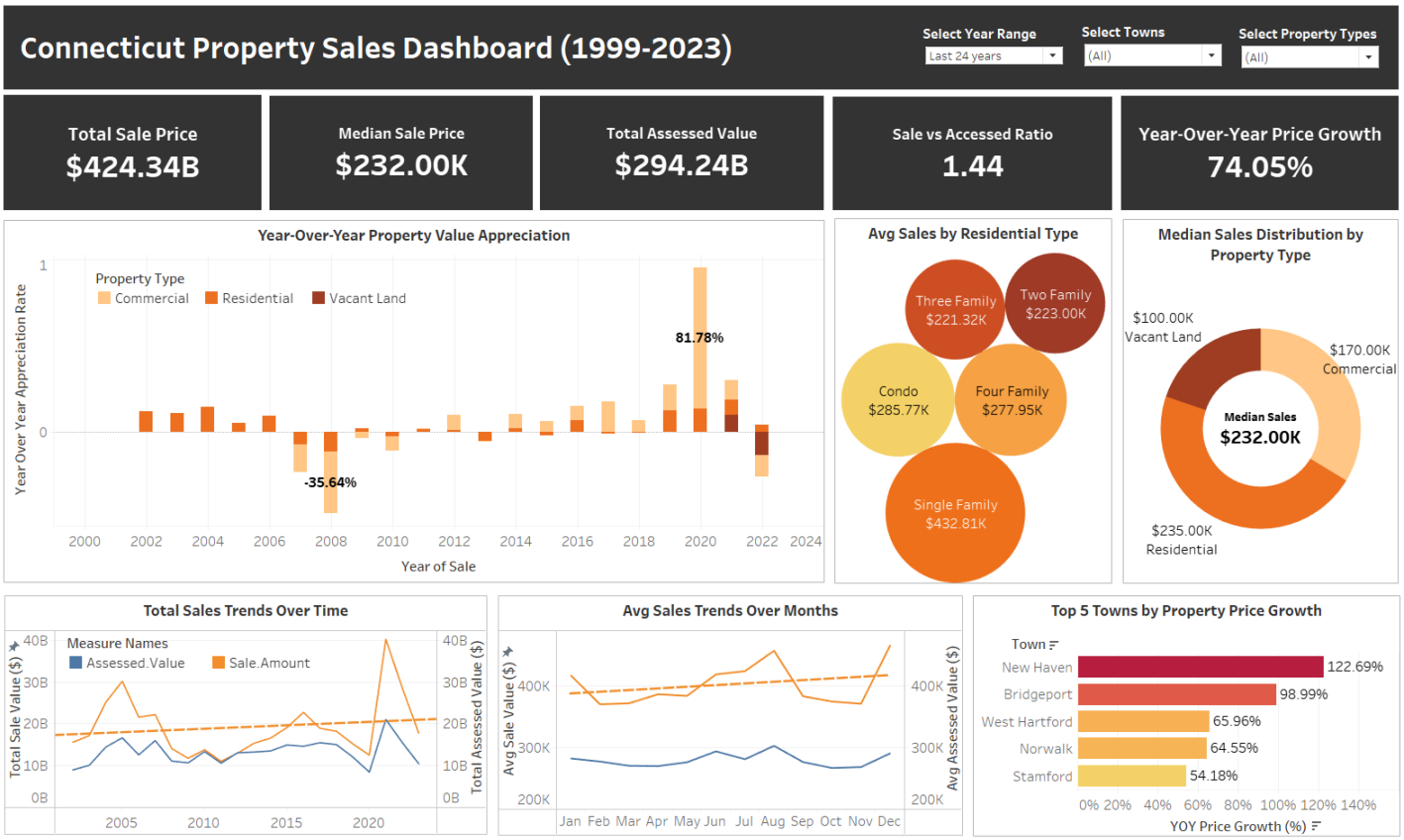
Final Tableau Dashboard Snapshot: